

Bo Zhang

Research Fellow | Department of Electrical and Computer Engineering
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Research profile

I am a Research Fellow in the Department of Electrical and Computer Engineering at the National University of Singapore (NUS). My research focuses on using metalines, metasurfaces and metamaterials to control antenna radiation through engineered electromagnetic boundary conditions.

I am interested in improving radiation-pattern roundness and wireless coverage in constrained environments. My current work also explores generative machine learning for the inverse design of electromagnetic structures, including high-permeability metamaterials.

Education

2025 | Ph.D.

National University of Singapore

2020 | M.S.

National University of Singapore

2019 | B.Eng. in Electromagnetic Field and Wireless Technology

University of Electronic Science and Technology of China

Research appointment

2025-present | Research Fellow

Department of Electrical and Computer Engineering, National University of Singapore

Awards and honors

2025 | First Prize, Student Paper Contest

IEEE Asia-Pacific Conference on Antennas and Propagation (APCAP)

2022 | Excellent Award

The 3rd NUS EM & Microwave Student Symposium

2021 | Honorable Mention

IEEE AP-S Student Paper Competition

Academic service

Reviewer: IEEE Transactions on Antennas and Propagation, IEEE APCAP and RFAT.

Member: IEEE RFID Singapore Chapter.

ICEAA 2026 contribution

cWGAN-Synthesized Unit Cells for Compact Near-Field Metasurface Beam Tilting.

ICEAA-IEEE APWC 2026, Paper 372.

Selected publications

Journal articles

Bo Zhang, Zhi Ning Chen. **Compact Near-Field Metasurface for 2-D Antenna Beam Tilting**. IEEE Transactions on Antennas and Propagation, vol. 74, no. 7, pp. 6374-6382, 2026.
[IEEE Xplore](#) | DOI: 10.1109/tap.2026.3685888

Bo Zhang, Zhi Ning Chen. **Guided-Wave Suppression in Uniaxial Anisotropic Metamaterial Slab for Radiation Pattern Control of Covered Dipoles**. IEEE Transactions on Antennas and Propagation, vol. 73, no. 2, pp. 952-961, 2025.
[IEEE Xplore](#) | DOI: 10.1109/tap.2024.3474799

Bo Zhang, Zhi Ning Chen, Yucong Zhou, Qun Lou, Jiahao Wang, Koen Mouthaan. **Improving Radiation Pattern Roundness of Henge-Like Metaring-Loaded Monopoles Above a Finite Ground for MIMO Systems**. Electromagnetic Science, vol. 2, no. 2, pp. 1-12, 2024.
[IEEE Xplore](#) | DOI: 10.23919/emsci.2024.0018

Zhi Ning Chen, Xianming Qing, Wei Liu, Huiwen Sheng, Teng Li, **Bo Zhang**. **Toward Metantennas: Metasurface Antennas Shaping Wireless Communications**. IEEE Communications Magazine, vol. 62, no. 8, pp. 94-99, 2024.
[IEEE Xplore](#) | DOI: 10.1109/mcom.001.2400047

Conference papers

Bo Zhang, Zhi Ning Chen. **Drude Model-Based Metamaterial Cover for Wideband Radiation Pattern Roundness Improvement**. 2025 IEEE 13th Asia-Pacific Conference on Antennas and Propagation (APCAP), pp. 62-63, 2025.
[IEEE Xplore](#) | DOI: 10.1109/apcap65837.2025.11210791

Zhi Ning Chen, **Bo Zhang**, Huiwen Sheng. **Localizing Radiation for Radiation Pattern Manipulation of Monopole Antennas With Finite-Sized Ground Planes**. 2025 19th European Conference on Antennas and Propagation (EuCAP), pp. 1-3, 2025.
[IEEE Xplore](#) | DOI: 10.23919/eucap63536.2025.10999378

Bo Zhang, Zhi Ning Chen, Yucong Zhou. **360° Single/Dual-Beam Steering of an EBG Encircled Monopole Antenna by Controlling Ground Surface Currents**. 2023 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (USNC-URSI), pp. 119-120, 2023.
[IEEE Xplore](#) | DOI: 10.1109/usnc-ursi52151.2023.10238095

Bo Zhang, Zhi Ning Chen. **Radiation Pattern Roundness Improvement of Off-center Monopole Antenna Using Electromagnetic Band-gap (EBG) Structure**. 2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (APS/URSI), pp. 1313-1314, 2021.
[IEEE Xplore](#) | DOI: 10.1109/aps/ursi47566.2021.9704411